Deriving Derivatives

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$\underline{f(x) = x^2}$$

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}$$

$$(x+h)^2 = 1(x)^2(h)^0 + 2(x)^1(h)^1 + 1(x)^0(h)^2$$
$$x^2 + 2xh + h^2$$

$$\lim_{h\to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h} \longrightarrow \lim_{h\to 0} \frac{2xh + h^2}{h}$$

$$\lim_{h\to 0} \frac{2x\cancel{h}}{\cancel{h}} + \underbrace{\lim_{h\to 0} h}$$

$$f'(x) = 2x + \lim_{h\to 0} h$$

$$\boxed{f'(x) = 2x}$$

$$\underline{g(x) = x^3} \qquad g'(x) = \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h}$$

$$(x+h)^3 = 1(x)^3(h)^0 + 3(x)^2(h)^1 + 3(x)^1(h)^2 + 1(x)^0(h)^3$$
$$x^3 + 3x^2h + 3xh^2 + h^3$$

$$\lim_{h\to 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x^3}}{h}$$

$$\lim_{h\to 0} \frac{3x^2\cancel{h}}{\cancel{h}} + \underbrace{\lim_{h\to 0} \frac{3xh^2}{h}}_{\lim_{h\to 0} 3xh \,\downarrow\, 0} + \underbrace{\lim_{h\to 0} \frac{h^3}{h}}_{\lim_{h\to 0} h^2 \,\downarrow\, 0}$$

$$\boxed{g'(x) = 3x^2}$$

$$\underline{h(x) = x^4} \qquad h'(x) = \frac{(x+h)^4 - x^4}{h}$$

$$(x+h)^4 = 1(x)^4(h)^0 + 4(x)^3(h)^1 + 6(x)^2(h)^2 + 4(x)^1(h)^3 + 1(h)^4$$
$$x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$$

$$\lim_{h\to 0} \frac{\cancel{x^4} + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - \cancel{x^4}}{h}$$

$$\lim_{h\to 0} \frac{4x^3\cancel{h}}{\cancel{h}} + \underbrace{\lim_{h\to 0} 6x^2h}_{0} + \underbrace{\lim_{h\to 0} 4xh^2}_{0} + \underbrace{\lim_{h\to 0} h^3}_{0}$$

$$\boxed{h'(x) = 4x^3}$$

Binomial expansion

0				1			
1			1		1		
2		1		2		1	
3	1		3		3		1
4		1		4		6		4		1
5	1		5		10		10		5		1
6	1		6		15		20		15		6		1

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Terms

$$r^{th} \text{ term} = x^r \cdot \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

$$\underline{d(x) = x^n} \qquad d'(x) = \lim_{h\to 0} \frac{(x-h)^n - x^n}{h}$$

$$(x-h)^n = x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \dots + h^n$$

$$d'(x) = \lim_{h\to 0} \frac{\cancel{x^n} + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \dots + h^n - \cancel{x^n}}{h}$$

$$d'(x) = \lim_{h\to 0} \frac{nx^{n-1}\cancel{h}}{\cancel{h}} + \underbrace{\lim_{h\to 0} \frac{n(n-1)}{2!}x^{n-2}h}_{0} + \dots \underbrace{\lim_{h\to 0} h^{n-1}}_{0}$$

$$\boxed{d'(x) = nx^{n-1}}$$

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